C Programming Lab Assignments

List of Problem Statements and Submission Dates

S.No	Problem Statement	Submission Date
6	**Linear Search** Write a C program to read *n* elements into an array and a search key from the user. Check each element one by one. If the key is found, display its index position; otherwise display "Element not found". Also display the number of comparisons made.	22/10/2026
7	**Binary Search** Write a C program to read *n* elements in sorted (ascending) order and a search key. Use the binary search technique, repeatedly halving the search range around the middle element, to find the key. Display its index if found, or "Element not found" otherwise.	22/10/2026
8	**Array Traversal and Sum using Functions** Write a C program with separate user-defined functions to: a) Read *n* elements into an array. b) Traverse the array and display every element with its index. c) Calculate and return the sum of all elements. Call these functions from `main()` and display the sum in a formatted manner.	22/10/2026
9	**Array Insertion and Deletion** Write a menu-driven C program to perform the following operations on an array: a) **Insert** a new element at a given index position by shifting the later elements one place to the right. b) **Delete** the element at a specific index position by shifting the later elements one place to the left. Check that the index is valid and that the array is not full before inserting. Display the array before and after each operation.	22/10/2026
10	**Recursive Functions** Write a C program using recursion to: a) Find the **factorial** of a given non-negative number *n*, where n! = n × (n - 1)! and 0! = 1. b) Find the **sum of array elements**, where the sum of *n* elements is the last element plus the sum of the first *n* - 1 elements. Clearly identify the base case and the recursive case in each function.	22/10/2026